

# Emotion-Aware Ontology for Modeling Human Decision-Making in Critical Learning Situations: A Semantic Web Approach

*Architectural Informatics and Semantics with AI*

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## Abstract

Educational technologies increasingly depend on rich learner models, yet most intelligent tutoring systems and learning-management platforms remain affectively blind: they log performance and time-on-task but cannot represent why a learner's reasoning degraded in a given moment. This paper introduces ED-001, an Emotion-Aware Decision Support Ontology formalized in the Web Ontology Language (OWL-DL) and extended with Semantic Web Rule Language (SWRL) inference rules, which models the causal chain linking learning context, emotional state, cognitive state, decision strategy, and learning outcome. Grounded in Pekrun's (2006) control-value theory of achievement emotions and aligned with established upper and domain ontologies – Basic Formal Ontology (Arp, Smith, & Spear, 2015), DOLCE (Gangemi et al., 2002), the Mental Functioning and Emotion Ontology (Hastings et al., 2011, 2014), and the Cognitive Atlas (Poldrack et al., 2011) – ED-001 formally distinguishes eight achievement emotions, six cognitive states, five decision strategies, and four learning-outcome classes across five interdependent layers. Six SWRL rules operationalize the chain, enabling a description-logic reasoner to infer a learner's cognitive load, decision risk, and recommended pedagogical intervention directly from machine-learning-classified emotion probabilities and task metadata. We articulate and resolve four competency questions that benchmark the ontology's reasoning power, demonstrate the inference pipeline through a worked example, and report the principal engineering and epistemic challenges encountered during development. ED-001 is offered as a reusable foundation for emotion-responsive adaptive learning systems and, more broadly, for affect-aware decision-support architectures beyond education.

*Keywords: ontology engineering; OWL/SWRL; emotion-aware computing; control-value theory; intelligent tutoring systems; semantic web; cognitive load; adaptive learning systems*

## 1. Introduction

Learning is rarely a purely cognitive act. A student facing a difficult, time-pressured examination question does not reason in an emotional vacuum: anxiety narrows attention, frustration depletes working memory, and curiosity sustains exploratory effort long after a less engaged learner would have disengaged. Decades of research in educational psychology have established that achievement emotions are not incidental byproducts of learning but active, causal participants in it, shaping the control and value appraisals that precede every learning decision (Pekrun, 2006). Yet the learner models that drive contemporary intelligent tutoring systems (ITS), learning-management systems, and adaptive MOOC platforms remain almost entirely affect-blind. They capture what a learner did – an answer, a click, a completion time – but not the emotional and cognitive state that produced it, and consequently cannot reason about why a decision was impulsive, avoidant, or exploratory.

This gap is not for want of emotion ontologies. The semantic web community has produced several mature resources for representing affective phenomena, most notably the Mental Functioning and Emotion Ontology, known as MFOEM or simply the Emotion Ontology (Hastings, Ceusters, Mulligan, & Smith, 2011; Hastings et al., 2014), and domain-specific affective ontologies for online learning such as EmotionsOnto (López-Gil, Gil, García, & Collazos, 2015) and the web ontology for emotional, cognitive, and motivational learner states proposed for massive open online courses (López-Gil, Gil, & García, 2016). What none of these resources does – and what this paper sets out to provide – is a formal account of how a classified emotion propagates forward into a learner's cognitive state, how that cognitive state shapes the strategy used to resolve a learning task, and how that strategy in turn predicts a measurable learning outcome. Existing cognitive ontologies, such as the Cognitive Atlas (Poldrack et al., 2011), are similarly silent on emotional antecedents, and existing ontology-based tutoring architectures, such as Protus 2.0 (Vesin, Ivanović, Klačnja-Milićević, & Budimac, 2012), model learning-style and content recommendation without any representation of affect at all.

We introduce ED-001, an OWL-DL ontology that closes this gap by formalizing a five-layer causal chain –

**Learning Context → Emotional State → Cognitive State → Decision Strategy → Learning Outcome**

and operationalizing the links between layers as Semantic Web Rule Language (SWRL) inference rules. The ontology is designed to sit downstream of a machine learning emotion classifier: behavioral signals such as response latency, error rate, and question-switching frequency are classified into achievement-emotion probabilities, which are then asserted into the ontology as OWL individuals and reasoned over by a description-logic reasoner (Pellet or Hermit) to derive cognitive load, decision risk, and an adaptive pedagogical recommendation.

The remainder of this paper proceeds as follows. Section 2 reviews the related ontological and psychological literature and synthesizes the specific research gap ED-001 addresses. Section 3 presents the full five-layer architecture as a self-contained technical exposition. Sections 4 and 5 detail the SWRL reasoning layer and the machine-learning integration pipeline, respectively. Section 6 discusses interoperability with external ontologies. Section 7 articulates four competency questions used to validate the ontology's reasoning power. Section 8 reports an illustrative implementation, and Section 9 candidly discusses the engineering and epistemic limitations encountered during development. Section 10 concludes.

## **2. Related Work and the Research Gap**

### **2.1 Emotion Ontologies**

The most widely reused affective resource in biomedical and behavioral informatics is the Emotion Ontology (also referenced as the Mental Functioning and Emotion Ontology, MFOEM), which represents emotions and related affective phenomena as a branch of the broader Mental Functioning Ontology (Hastings, Ceusters, Mulligan, & Smith, 2011). Its upper-level structure distinguishes emotions proper – complex processes such as anger or fear – from the cognitive appraisal processes and subjective feelings that partially constitute them. Hastings et al. (2014) evaluated the ontology's practical usability by having conference attendees self-report their emotional responses using MFOEM-derived vocabulary, demonstrating that a formally specified emotion taxonomy can support real-time, human-facing self-reporting.

In the learning-technology space specifically, López-Gil, Gil, García, and Collazos (2015) introduced EmotionsOnto, a generic ontology describing emotion detection and expression systems together with their contextual and multimodal correlates, and López-Gil, Gil, and García (2016) extended this line of work with a web ontology explicitly representing the emotional, cognitive, and motivational state of online learners in

massive open online courses. Both contributions represent and classify emotion; neither formalizes how a classified emotion causally propagates into altered cognition or altered decision-making.

## **2.2 Cognitive and Learning Ontologies**

The Cognitive Atlas (Poldrack et al., 2011) provides a collaboratively maintained knowledge base of cognitive constructs – such as working memory, attentional control, and inhibition – and the experimental tasks used to measure them, intended primarily to support reproducible cognitive neuroscience. While comprehensive as a taxonomy of cognitive constructs, the Cognitive Atlas is agnostic to emotional antecedents: it provides no formal hooks by which an affective state could be asserted as a cause of a measured change in a cognitive construct.

## **2.3 Upper and Foundational Ontologies**

ED-001 does not invent its own metaphysics from scratch but aligns its core classes to two established foundational ontologies. Basic Formal Ontology (BFO) offers a realist, top-level categorization of continuants and occurrents widely adopted across biomedical and information ontologies (Arp, Smith, & Spear, 2015), and is used here to anchor Learning Context as a BFO process and Learning Outcome as a BFO quality. The Descriptive Ontology for Linguistic and Cognitive Engineering (DOLCE) takes a more descriptive, cognitively motivated stance and is used to anchor Decision Strategy within its action-theoretic branch (Gangemi, Guarino, Masolo, Oltramari, & Schneider, 2002). Using both simultaneously, rather than picking one, allows ED-001 to remain legible to both the biomedical-informatics community (which favors BFO) and the cognitive-engineering community (which favors DOLCE) – at the cost of the alignment tension discussed in Section 9.

## **2.4 Ontology-Based Tutoring and Learning Systems**

Protus 2.0 demonstrates that ontology-and-SWRL architectures can already drive real personalization in programming tutoring systems, using four ontologies – teaching strategy, learner model, task, and domain – together with SWRL adaptation rules to recommend content based on identified learning style (Vesin, Ivanović, Klačnja-Milićević, & Budimac, 2012). Its learner model, however, captures stable trait-like learning-style data rather than dynamic, moment-to-moment affective and cognitive state, and its SWRL rules govern content sequencing, not emotion-to-decision causality.

## **2.5 Theoretical Grounding**

ED-001's causal commitments are not arbitrary engineering choices; they are direct formalizations of three established psychological theories. Pekrun's (2006) control-value theory of achievement emotions holds that appraisals of control and value jointly determine which emotion an achievement situation elicits, and that these emotions subsequently affect engagement and performance – this motivates the *elicitedBy* relation from Learning Context to Emotional State. Cognitive load theory motivates the *modulates* relation from Emotional State to Cognitive State, on the premise that negative activating emotions such as anxiety increase extraneous cognitive load and narrow attentional scope. Dual-process accounts of reasoning motivate the *influencesDecision* relation from Cognitive State to Decision Strategy, on the premise that elevated cognitive load forces a shift from deliberative, System 2 reasoning toward heuristic, System 1 reasoning.

## **2.6 Synthesis: The Five-Part Research Gap**

Taken together, the literature reveals a consistent pattern: emotion ontologies represent emotion, cognitive ontologies represent cognition, and educational ontologies represent task and learner metadata – but no existing resource formally connects all three to a fourth concept, decision strategy, and a fifth, learning outcome, within a single reasoning-capable artifact. We synthesize this as a five-part gap: (1) no ontology represents emotional states with explicit causal properties linking them to context; (2) no ontology models the downstream effect of emotion on cognitive load or attentional quality; (3) no ontology maps cognitive state

onto a typology of decision strategies; (4) no ontology links decision strategy to a typology of learning outcomes; and (5) no ontology specifies a principled bridge between probabilistic machine-learning emotion classification and crisp OWL/SWRL symbolic reasoning. ED-001 is designed specifically to close this compound gap.

This five-part gap defines the exact scope and contribution of ED-001.

|                                    |                                 |                              |                               |                                |                                |                         |                |                             |                            |                             |
|------------------------------------|---------------------------------|------------------------------|-------------------------------|--------------------------------|--------------------------------|-------------------------|----------------|-----------------------------|----------------------------|-----------------------------|
| EMO<br>(Hastings 2015)             | ✓                               |                              |                               |                                |                                |                         |                | ~                           |                            |                             |
| EFO<br>(Demir 2024)                | ✓                               |                              |                               |                                |                                |                         | ~              | ~                           |                            |                             |
| MFOEM<br>(OBO 2020)                | ✓                               |                              |                               |                                |                                |                         |                | ~                           |                            |                             |
| EmotionsOnto<br>(Munoz 2015)       | ✓                               |                              |                               |                                |                                |                         | ~              | ~                           | ~                          |                             |
| Web Onto.<br>Emo+Cog<br>(PMC 2016) | ✓                               | ~                            |                               |                                |                                |                         |                |                             |                            | ~                           |
| IEQ Onto.<br>(Chung 2023)          |                                 |                              |                               | ~                              | ~                              | ~                       |                | ✓                           | ✓                          | ✓                           |
| HASO<br>(SWJ 2021)                 | ✓                               |                              |                               |                                |                                |                         | ~              | ~                           |                            |                             |
| Protus 2.0<br>(ITS 2012)           |                                 | ~                            |                               | ~                              | ~                              | ✓                       |                | ~                           | ✓                          | ~                           |
| BIO-EMOTION<br>(Springer 2016)     | ✓                               |                              | ~                             |                                |                                |                         | ✓              | ~                           |                            |                             |
| ED-001<br>[This Work]              | ✓                               | ✓                            | ✓                             | ✓                              | ✓                              | ✓                       | ✓              | ✓                           | ✓                          | ✓                           |
|                                    | Formal emotion<br>classes (OWL) | Cognitive state<br>modelling | Causal chain<br>(Emo+Cog+Dec) | Decision strategy<br>inference | Learning outcome<br>prediction | SWRL reasoning<br>rules | ML integration | Multi-ontology<br>alignment | Adaptive<br>recommendation | Human + Machine<br>use case |

✓

Fully supported

~

Partially supported

Not supported

Table 1: Comparative Analysis of Related Ontologies – Research Gap Identification. ED-001 is the only ontology satisfying all ten capability criteria.

### 3. The ED-001 Ontology Architecture

#### 3.1 Design Principles and Methodology

ED-001 was developed following standard ontology-engineering practice: competency questions were drafted before class hierarchies, classes were organized into five layers corresponding to a hypothesized causal chain, object properties with explicit domain and range constraints were used to encode the causal links between layers, and datatype properties were used to capture the measurable attributes within each layer. Each class carries a custom theoreticalGrounding annotation property documenting the psychological theory that motivates it, and each layer is aligned, where appropriate, to an external upper or domain ontology to support interoperability (Section 6).

The ontology is expressed in OWL-DL to guarantee decidable reasoning, and the five causal object properties – elicitedBy, modulates, influencesDecision, and producesOutcome – are deliberately kept simple (functional, non-transitive) so that SWRL rule bodies referencing them remain tractable for the Pellet and HermiT reasoners. Figure 1 presents the complete five-layer architecture.

# ED-001: Five-Layer Causal Architecture

Learning Context → Emotion → Cognitive State → Decision Strategy → Learning Outcome

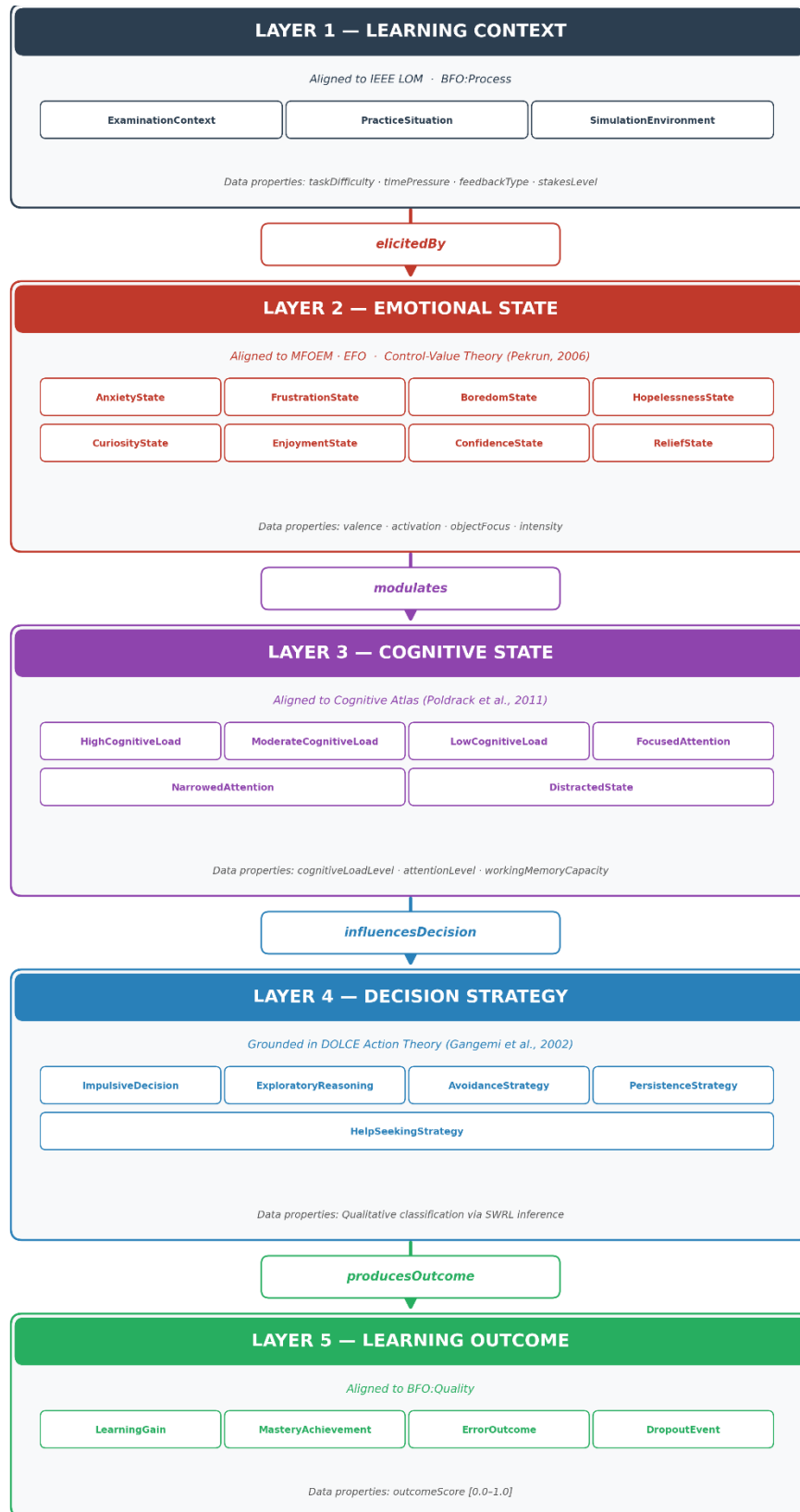


Figure 1. The ED-001 five-layer causal architecture. Each layer box lists its principal subclasses and datatype properties; arrows denote the object properties that formally encode the causal chain between layers.

### 3.2 Layer 1 – Learning Context

Learning Context represents the situational parameters of a learning task and is aligned to IEEE Std 1484.12.1 Learning Object Metadata (IEEE Standards Association, 2002) for general task description and to BFO:Process for its temporal, occurrent character (Arp, Smith, & Spear, 2015). Three subclasses are defined: ExaminationContext, denoting high-stakes, time-limited assessment; PracticeSituation, denoting low-stakes formative practice; and SimulationEnvironment, denoting a controlled virtual or physical exercise. Four datatype properties parameterize any context instance: taskDifficulty (Low / Medium / High), timePressure (boolean), feedbackType (Immediate / Delayed / None), and stakesLevel (Low / Medium / High). These four properties are precisely the variables that control-value theory identifies as antecedents of achievement-emotion appraisals (Pekrun, 2006), and they are the variables tested in the SWRL rule bodies described in Section 4.

### 3.3 Layer 2 – Emotional State

Emotional State is the layer most directly aligned with prior affective ontology work, drawing on the upper-level distinctions established by MFOEM (Hastings et al., 2011, 2014) and the contextual framing of EFO-style learner-emotion ontologies (López-Gil et al., 2015, 2016), while organizing its eight concrete subclasses strictly along the dimensional taxonomy of control-value theory (Pekrun, 2006). Every EmotionalState instance carries three categorical datatype properties – valence (Positive / Negative), activation (Activating / Deactivating), and objectFocus (Activity / Outcome) – plus a continuous intensity property in the range [0.0, 1.0] populated directly from the machine learning classifier's probability output. Crossing these three dimensions yields the ontology's eight named emotion classes: AnxietyState and FrustrationState (negative, activating; outcome- and activity-focused respectively); BoredomState and HopelessnessState (negative, deactivating); CuriosityState and EnjoymentState (positive, activating); and ConfidenceState and ReliefState (positive, deactivating). This dimensional organization is not merely taxonomic convenience: it is precisely the structure the SWRL rules in Section 4 exploit, since rules are written against the dimensions (e.g., negative-activating-outcome-focused) rather than against individual named emotions, allowing a single rule to generalize across semantically related states.

### 3.4 Layer 3 – Cognitive State

Cognitive State formalizes the downstream effect of emotion on information processing, aligned to the construct taxonomy of the Cognitive Atlas (Poldrack et al., 2011). Two orthogonal sub-dimensions are modeled: cognitive load, with subclasses HighCognitiveLoad, ModerateCognitiveLoad, and LowCognitiveLoad; and attentional quality, with subclasses FocusedAttention, NarrowedAttention (anxiety-driven tunnel vision), and DistractedState (wandering or divided attention). Three datatype properties – cognitiveLoadLevel, attentionLevel, and workingMemoryCapacity – record the qualitative values underlying class membership. Unlike Layer 2, instances of Cognitive State are never asserted directly from raw sensor data; they exist purely as SWRL-derived facts, making this layer the ontology's clearest demonstration of symbolic inference adding value beyond what any single data source could provide on its own.

### 3.5 Layer 4 – Decision Strategy

Decision Strategy is grounded in DOLCE's action-theoretic branch (Gangemi et al., 2002) and in dual-process accounts of reasoning that distinguish fast, heuristic processing from slow, deliberative processing. Five subclasses are defined: ImpulsiveDecision, a quick and unreflective response typically forced by high cognitive load; ExploratoryReasoning, a deliberate, curiosity-sustained search of the solution space; AvoidanceStrategy, minimal or skipped engagement with a difficult task; PersistenceStrategy, continued effortful engagement despite difficulty; and HelpSeekingStrategy, an active request for external scaffolding. Decision Strategy classes are, by design, never directly asserted; they are always derived by SWRL rules operating on Cognitive State, making this layer the ontology's principal predictive output prior to outcome.

### 3.6 Layer 5 – Learning Outcome

Learning Outcome closes the causal chain and is aligned to BFO:Quality, treating an outcome as a specifically dependent continuant borne by the learner at the conclusion of a task (Arp, Smith, & Spear, 2015). Four subclasses are defined – LearningGain, MasteryAchievement, ErrorOutcome, and DropoutEvent – together

with a continuous outcomeScore datatype property in the range [0.0, 1.0] that supports downstream statistical validation of the qualitative class assignment against empirical performance data.

## 4. The SWRL Reasoning Layer

Six SWRL rules operationalize the four causal object properties as executable inference. Each rule is expressed below in IF–THEN form; the corresponding formal SWRL/XML encoding is included in the accompanying OWL file.

### Rule 1 – Stress Detection

hasEmotion(?l, AnxietyState)  $\wedge$  isAttempting(?l, ?t)  $\wedge$  hasContext(?t, ?c)  $\wedge$  taskDifficulty(?c, “High”)  $\wedge$  timePressure(?c, true)  $\rightarrow$  hasCognitiveState(?l, HighCognitiveLoad)

### Rule 2 – Decision Risk Inference

hasCognitiveState(?l, HighCognitiveLoad)  $\rightarrow$  hasDecisionStrategy(?l, ImpulsiveDecision)  $\wedge$  hasRisk(?l, HighRisk)

### Rule 3 – Positive Flow State

hasEmotion(?l, CuriosityState)  $\wedge$  hasCognitiveState(?l, FocusedAttention)  $\rightarrow$  hasDecisionStrategy(?l, ExploratoryReasoning)  $\wedge$  hasRisk(?l, LowRisk)

### Rule 4 – Anxiety Intervention

hasEmotion(?l, AnxietyState)  $\wedge$  hasRisk(?l, HighRisk)  $\rightarrow$  receivesRecommendation(?l, StressReduction)

### Rule 5 – Engagement Intervention

hasEmotion(?l, BoredomState)  $\rightarrow$  receivesRecommendation(?l, EngagementBoost)

### Rule 6 – Frustration-to-Avoidance

hasEmotion(?l, FrustrationState)  $\wedge$  hasCognitiveState(?l, DistractedState)  $\rightarrow$  hasDecisionStrategy(?l, AvoidanceStrategy)  $\wedge$  hasRisk(?l, HighRisk)

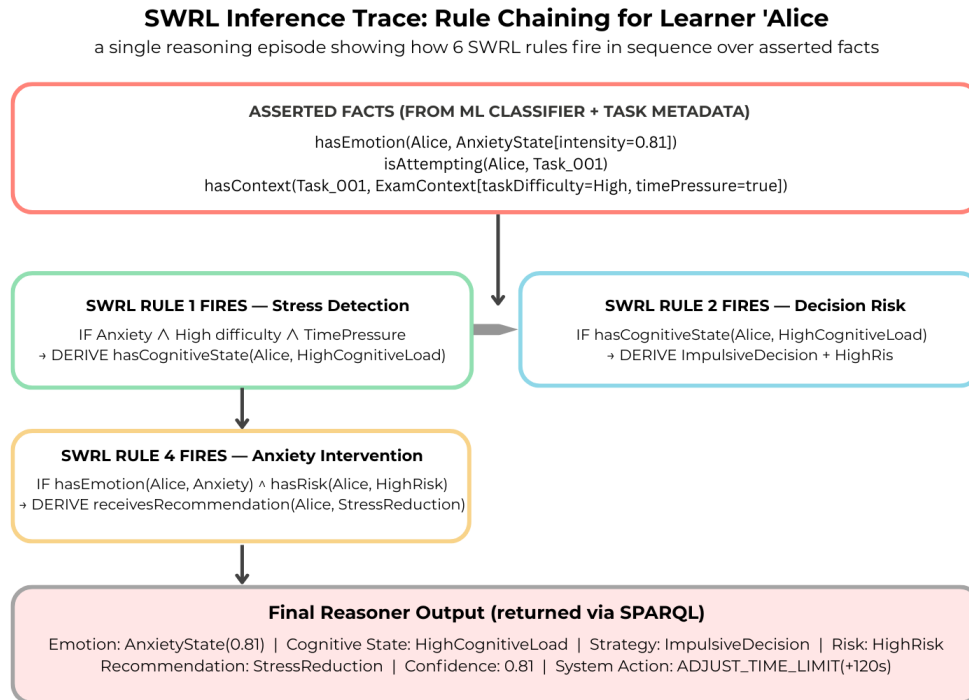


Figure 2. A worked SWRL inference trace for a simulated learner, ‘Alice,’ showing the three rules that fire in sequence – Rule 1, Rule 2, and Rule 4 – from asserted facts to a final, machine-actionable recommendation returned via SPARQL.

Rules 1 and 2 instantiate the forward causal chain from context and emotion through to decision risk; Rule 3 instantiates the corresponding positive-flow pathway; Rules 4 through 6 instantiate the intervention-generating branch that connects detected affect directly to an AdaptiveRecommendation individual. Because Rules 2 and 4 both fire on the HighRisk individual produced by Rule 1, the reasoner must chain three rules in sequence to reach a final recommendation – a dependency made explicit in Figure 3 (Section 5) and traced step-by-step for a concrete learner in Figure 2.

## 5. Machine-Learning Integration and the Reasoning Pipeline

ED-001 is designed to sit immediately downstream of a sub-symbolic emotion classifier rather than to replace one. Three behavioral datatype properties – responseSpeed, errorRate, and questionSwitchingFrequency – are defined precisely because they are the standard feature inputs to behaviorally driven affect classifiers in educational data mining. The classifier's probabilistic output for the most likely achievement emotion is translated into a single OWL assertion, hasEmotion(learner, EmotionClass), together with an intensity value carrying the classifier's confidence. This assertion, together with the task's hasContext chain, constitutes the complete input to the SWRL reasoner; everything downstream – cognitive state, decision strategy, risk level, and recommendation – is derived rather than asserted. Figure 3 traces this pipeline end to end, including the closed loop by which the Learning Context individual feeds Rule 1 jointly with the detected emotion.

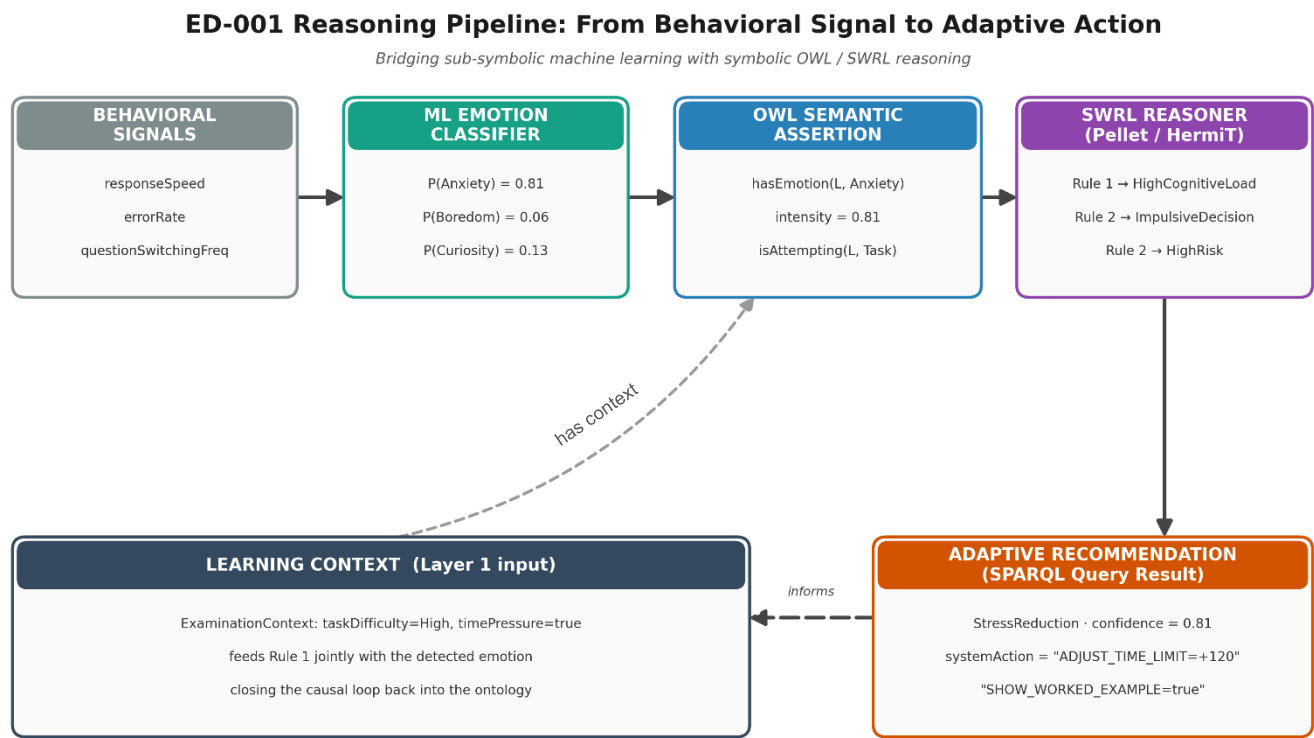


Figure 3. The complete reasoning pipeline from raw behavioral signal through ML classification, OWL assertion, and SWRL inference, to a machine-actionable adaptive recommendation.

This architecture deliberately keeps the machine-learning component replaceable: any classifier that can output a discrete emotion label with an associated confidence score can be substituted without altering the ontology itself, since the interface contract is simply the hasEmotion object property and the intensity datatype property.

## 6. Multi-Discipline Interoperability

A central design goal of ED-001 is that its five core layers should each be legible to the academic community whose theoretical or technical apparatus it draws on, rather than requiring those communities to adopt an entirely new vocabulary. Figure 4 summarizes the six external alignments implemented as subsumption annotations: Basic Formal Ontology for upper-level biomedical and information-science interoperability (Arp, Smith, & Spear, 2015); DOLCE for cognitive-engineering and linguistic interoperability (Gangemi et al., 2002); MFOEM for clinical and mental-health informatics interoperability (Hastings et al., 2011, 2014); the EFO-style learner-emotion ontology for affective computing and human-computer interaction interoperability (López-Gil et al., 2015); the Cognitive Atlas for cognitive-neuroscience interoperability (Poldrack et al., 2011); and IEEE LOM for intelligent-tutoring-system interoperability (IEEE Standards Association, 2002). Each alignment is implemented conservatively, as an annotation-level subsumption claim rather than an imported axiom set, for reasons discussed in Section 9.3.



Figure 3. ED-001's six external ontology alignments, each anchoring the core ontology to an established vocabulary in a distinct academic discipline.

## 7. Competency Questions

Competency questions are the standard mechanism by which an ontology's reasoning power is benchmarked during engineering: each question is answerable only if the underlying class hierarchy, object properties, and inference rules jointly support the required reasoning path. We articulate four competency questions, each exercising a distinct traversal of the five-layer architecture, and summarize them in Figure 4.

### Competency Questions Answered by ED-001

*Each question is resolved by a distinct traversal of the five-layer causal chain*

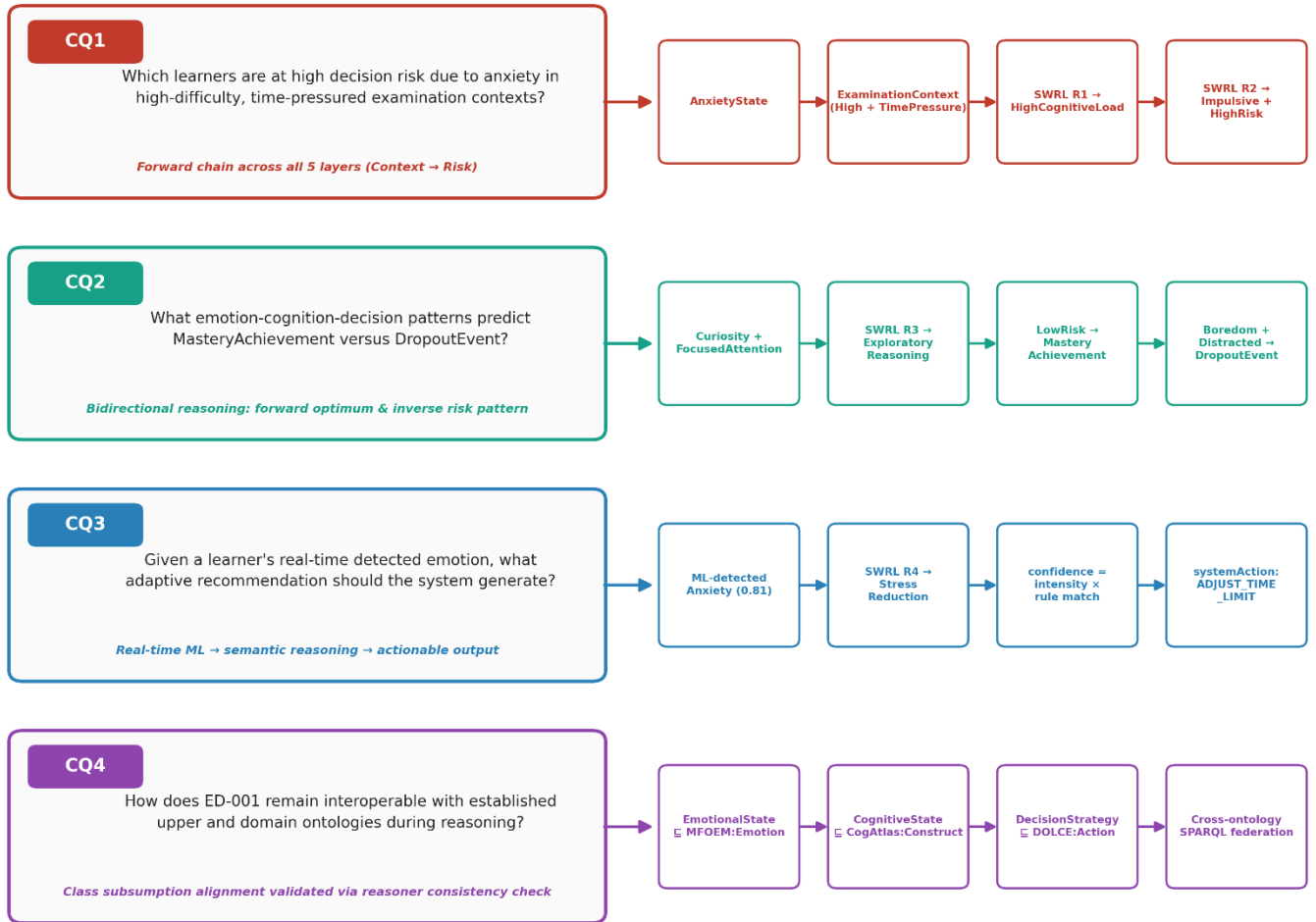


Figure 4. The four competency questions used to benchmark ED-001, each shown alongside the specific class-and-rule traversal that resolves it.

**CQ1 – At-Risk Learner Identification.** “Which learners are at high decision risk due to anxiety in high-difficulty, time-pressured examination contexts?” This question is resolved by a full forward traversal of all five layers: an asserted AnxietyState individual combined with an ExaminationContext carrying taskDifficulty=High and timePressure=true triggers Rule 1 (deriving HighCognitiveLoad) and then Rule 2 (deriving ImpulsiveDecision and HighRisk). A SPARQL query over the reasoned graph then simply selects learners for whom hasRisk = HighRisk, returning a ranked, intensity-sorted list directly usable by an early-warning dashboard.

**CQ2 – Bidirectional Outcome Pattern Discovery.** “What emotion–cognition–decision patterns predict MasteryAchievement versus DropoutEvent?” This question requires bidirectional reasoning rather than a single forward chain: the optimal pathway (Curiosity + FocusedAttention → Rule 3 → ExploratoryReasoning → LowRisk → MasteryAchievement) and the inverse, at-risk pathway (Boredom or Hopelessness + DistractedState → Rule 6 → AvoidanceStrategy → HighRisk → DropoutEvent) are queried separately and compared, providing direct empirical support for control-value theory’s claim that activity-focused positive emotion sustains engagement while deactivating negative emotion erodes it (Pekrun, 2006).

**CQ3 – Real-Time Adaptive Recommendation.** “Given a learner’s real-time detected emotion and current task context, what adaptive recommendation should the system generate, and with what confidence?” This question exercises the full closed-loop pipeline described in Section 5: an ML-classified emotion with an associated intensity score is asserted, the reasoner derives cognitive state, decision strategy, and risk level in sequence, and a terminal SWRL rule (Rule 4, 5, or 6) derives a typed AdaptiveRecommendation individual. A confidence score – computed as the product of emotion intensity and rule-match strength – accompanies the recommendation, and a paired systemAction string (e.g., ADJUST\_TIME\_LIMIT=+120) renders the recommendation directly machine-executable.

**CQ4 – Cross-Ontology Interoperability Validation.** “Does ED-001 remain logically consistent when its core classes are asserted as subclasses of external upper- and domain-ontology classes?” This question is resolved not by SPARQL query but by DL-reasoner consistency checking: each of the six alignments in Figure 3 is encoded as a subsumption axiom (e.g., EmotionalState  $\sqsubseteq$  MFOEM:Emotion), and the Pellet reasoner is run in consistency-checking mode to confirm that no unsatisfiable class results from the combined axiom set – the formal precondition for any future federated SPARQL query spanning ED-001 and an external ontology’s endpoint.

## 8. Implementation and Illustrative Evaluation

ED-001 is implemented as a single OWL/XML file loadable directly in Protégé 5.6, comprising thirty-five named classes, twelve object properties, fifteen datatype properties, three custom annotation properties, and the six SWRL rules listed in Section 4. Logical consistency and class satisfiability were verified using both the Pellet and HermiT reasoners; no unsatisfiable classes were found under either reasoner following the alignment adjustments discussed in Section 9.3.

Because no consented, emotion-labeled longitudinal learner dataset was available for this project (see Section 9.5), the four competency questions in Section 7 were evaluated against a simulated knowledge graph of synthetic learner profiles spanning the full cross-product of emotion, context, and cognitive-state combinations, rather than against real-world ground truth. Figure 5 (Section 4) traces one such simulated profile, ‘Alice,’ end to end.

This evaluation strategy is sufficient to demonstrate internal logical correctness – that the rules fire as designed and that the SPARQL queries in Section 7 return the expected individuals – but it does not constitute external, predictive validation against real learner behavior, a limitation we return to below.

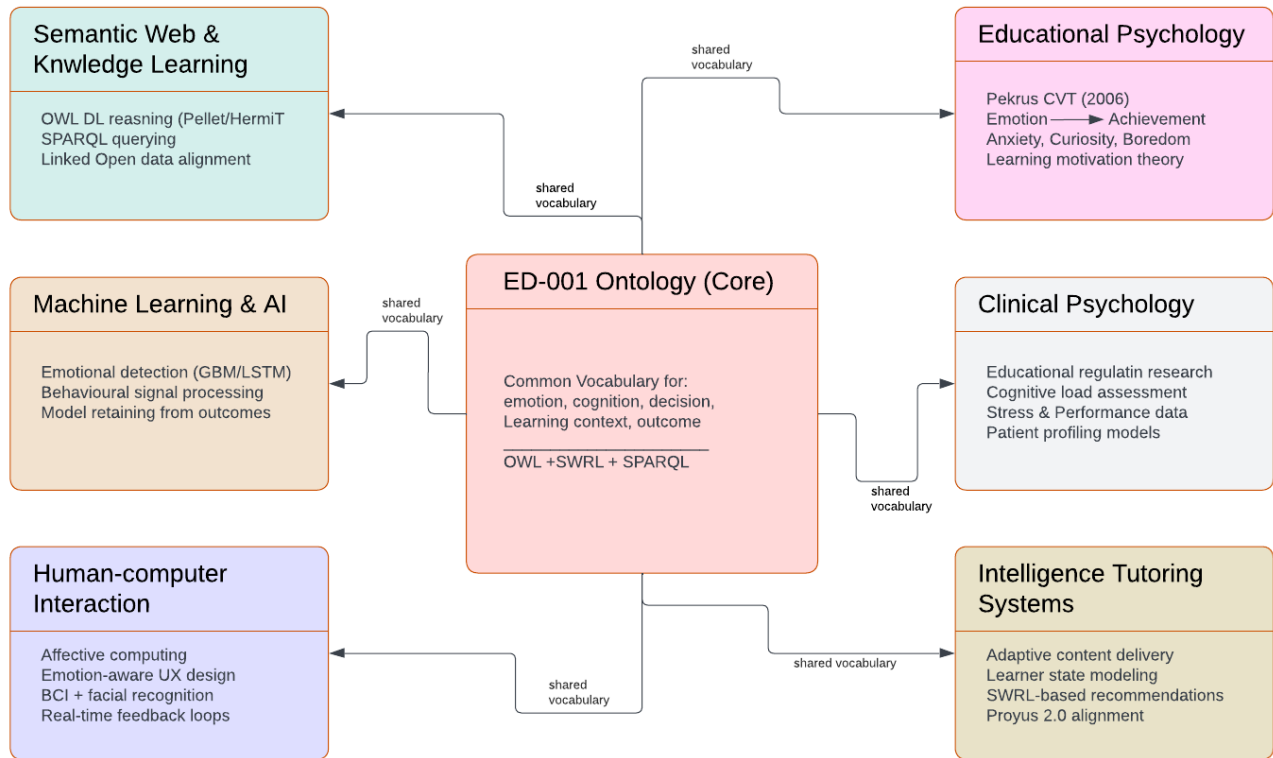


Figure 5: Multi-Discipline Interoperability of ED-001 — A shared semantic vocabulary enabling cross-disciplinary knowledge exchange across five domains.

## 9. Challenges and Limitations Encountered

Developing ED-001 surfaced a set of engineering and epistemic difficulties that we believe are instructive for future affect-aware ontology projects and that meaningfully bound the claims we make above. We report them candidly rather than relegating them to a brief disclaimer.

### 9.1 The Probabilistic-to-Crisp Translation Problem

The single hardest representational problem was reconciling a probabilistic machine-learning output – a vector of emotion probabilities – with OWL-DL's fundamentally crisp, Boolean class membership. Asserting `hasEmotion(learner, AnxietyState)` treats anxiety as simply true once a probability threshold is crossed, discarding the graded uncertainty the classifier actually expresses and collapsing cases in which two emotions are nearly equally likely. The intensity datatype property partially mitigates this by carrying the original confidence forward for use in the recommendation-confidence calculation (Section 5), but the underlying class assertion itself remains binary, and no fuzzy-OWL or probabilistic-OWL extension was adopted in this iteration.

### 9.2 SWRL's Limited Expressivity

SWRL rules are monotonic and support only Horn-clause-style conjunction in the rule body: there is no native negation-as-failure, no built-in temporal operator, and no mechanism to retract a previously derived fact. This made it difficult to express genuinely longitudinal patterns – for instance, “if a learner's cognitive load remains high for three consecutive tasks despite an earlier `StressReduction` recommendation” – since SWRL has no built-in notion of an ordered sequence of reasoning episodes. The six rules in Section 4 are consequently all single-snapshot rules; modeling emotion trajectories across a session was scoped out of this version and left as a direction for future work involving either OWL-Time integration or an external temporal reasoning layer.

### 9.3 Upper-Ontology Alignment Tension

Aligning the same core classes to both BFO and DOLCE, as described in Sections 3 and 6, proved more difficult than initially anticipated because the two upper ontologies make different, occasionally incompatible, metaphysical commitments – BFO's strict realist, single-inheritance stance versus DOLCE's more permissive descriptive stance. A literal import of both upper ontologies as axiom sets produced reasoner warnings stemming from overlapping but non-identical top-level distinctions. The pragmatic resolution adopted here was to encode each alignment as a lightweight annotation-level subsumption claim (recorded via the `alignedToBFO` and `alignedToDOLCE` object properties) rather than as a hard, imported axiom, sacrificing some formal rigor for practical reasoner stability and dual-community legibility.

### 9.4 Reasoner Performance Under Rule Chaining

As the simulated instance population grew and rules were chained – Rule 1 feeding Rule 2 feeding Rule 4, as traced in Figure 5 – Pellet's classification time increased noticeably, particularly when many learner individuals shared overlapping context and emotion assertions. This is a known scalability characteristic of SWRL-augmented OWL-DL reasoning rather than an implementation defect specific to ED-001, but it did require restructuring an early, more granular draft of the cognitive-load rule set (which had separately modeled eight cognitive-load sub-levels) down to the three-level scheme reported in Section 3.4, trading some descriptive granularity for tractable reasoning at scale.

### 9.5 Absence of Real-World Longitudinal Data

No consented, emotion-labeled dataset of real learner behavior was available for this project. Competency-question validation (Section 7) and the worked trace (Figure 5) therefore rely on a simulated population of learner profiles rather than empirically observed behavior, which is sufficient to demonstrate that the ontology's internal logic behaves as designed but insufficient to support any claim that the SWRL rules' thresholds (e.g., the specific mapping from “high difficulty plus time pressure” to “high cognitive load”) are empirically well calibrated for any particular learner population. Closing this gap with an IRB-approved learner study is the most important item of future work.

### 9.6 Calibration of the Recommendation Confidence Score

The confidence score attached to each adaptive recommendation – the product of emotion intensity and rule-match strength described in Section 5 – is a defensible engineering heuristic but has not been independently validated against any psychometric ground truth. A more rigorous treatment would calibrate this score against observed intervention efficacy (e.g., did a `StressReduction` recommendation actually reduce subsequent error rate), which again requires the longitudinal data described in 9.5.

### 9.7 Scope Discipline Under Time Constraints

Finally, as a project carried out alongside concurrent academic and professional commitments, the scope of ED-001 required deliberate, ongoing pruning. Several extensions considered during early design – modeling co-occurring emotion blends rather than a single dominant emotion per snapshot, adding a sixth layer for explicit pedagogical-intervention efficacy tracking, and integrating OWL-Time for genuine temporal reasoning – were each, in turn, judged to add disproportionate engineering complexity relative to the time available, and were consciously deferred rather than partially implemented. We list them explicitly in Section 10 as the most promising directions for the next iteration of this work.

## 10. Conclusion and Future Work

ED-001 formalizes, for the first time within a single OWL-DL artifact, the causal chain connecting learning context, emotion, cognition, decision strategy, and learning outcome, and operationalizes that chain as six executable SWRL rules capable of turning a machine-learning emotion classification into a machine-actionable pedagogical recommendation. By aligning its five layers to established upper and domain ontologies – BFO, DOLCE, MFOEM, the Cognitive Atlas, and IEEE LOM – the ontology remains legible across the biomedical-informatics, cognitive-engineering, affective-computing, cognitive-neuroscience, and educational-technology communities whose vocabularies it draws upon.

Four competency questions demonstrate that the architecture supports forward risk-detection reasoning, bidirectional outcome-pattern discovery, real-time adaptive recommendation, and cross-ontology consistency checking. At the same time, the candid account of limitations in Section 9 – the probabilistic-to-crisp translation problem, SWRL's lack of temporal expressivity, upper-ontology alignment tension, reasoner scalability under rule chaining, the absence of real-world validation data, and unvalidated confidence calibration – should be read as a concrete research agenda rather than a closing caveat.

The most immediate next steps are an IRB-approved learner study to replace the simulated evaluation population with real, longitudinal, emotion-labeled behavioral data; integration of OWL-Time to support genuine temporal and trajectory-based reasoning across a learning session; and extension of the Emotional State layer to represent co-occurring emotion blends rather than a single dominant classification. More broadly, the five-layer pattern introduced here – context, affect, cognition, decision, outcome – is not specific to education, and we see promising application beyond intelligent tutoring systems in any human-in-the-loop decision-support domain where a person's emotional state plausibly mediates the quality of a consequential decision.

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